Fact & Dimension Identification

NYC Ride-Hailing, Weather & EV Charging Access

Course	DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore
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1. Fact Tables

Fact Table	Type	Measures (additivity)	Degenerate Dim.
Fact_Trips Ride-hailing trips	Transaction	trip_miles, trip_time_sec, base_fare, tips, driver_pay, wait_min (all additive). Average wait = total wait / trips.	trip_id
Fact_Zone_Hourly_Snapshot Zone demand and weather	Periodic snapshot	trip_count, total_wait_min, total_driver_pay (additive). rainfall_mm (semi-additive: add across hours, not across zones, because one city reading repeats in every zone).	None
Fact_Charger_Coverage Charging station openings	Factless	No measures. Count rows to count stations.	None

2. Dimension Tables

Dimension	Natural Key	Surrogate Key	Key Attributes	SCD	Hierarchy
Dim_Zone	LocationID	zone_key	zone, borough, service_zone	Type 1	Borough to Zone
Dim_Station	station_id	station_key	network, ports, charger level, status	Type 2	None
Dim_Company	hvfhs_license_num	company_key	company name (Uber, Lyft, Via, Juno), base number	Type 1	None
Dim_Date	date	date_key	day, weekday, month, quarter, year	None	Year to Quarter to Month to Date
Dim_TimeOfDay	hour	time_key	hour, peak / off-peak, day part	None	None
Dim_Weather	weather_band	weather_key	rain level (none, light, heavy)	None	None

3. Which Dimensions Connect to Which Fact

Fact Table	Zone	Station	Company	Date	TimeOfDay	Weather
Fact_Trips	Yes (pickup and drop-off)	-	Yes	Yes	Yes	Yes
Fact_Zone_Hourly_Snapshot	Yes	-	-	Yes	Yes	Yes
Fact_Charger_Coverage	Yes	Yes	-	Yes	-	-

Notes: Dim_Zone is used twice in Fact_Trips (pickup zone and drop-off zone), so it plays two roles. Dim_Station is Type 2 so the warehouse keeps history when a station changes its network, ports or status. All measures follow the grain statements in the Grain Definition Document.